



NEXTLEAP PM FELLOWSHIP · SPOTIFY · GROWTH TEAM

# Off Repeat

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Spotify perfected recommendation. Discovery still stalls — because the gap was never accuracy. It's acceptance: a reason to press play on the unfamiliar.

*You have On Repeat. Meet Off Repeat.*

► **Live prototype + AI workflow** · [github.com/arjuncoolitr/spotify-discovery-engine](https://github.com/arjuncoolitr/spotify-discovery-engine)

Figures: Spotify reported MAU / Premium; repeat-listening & genre-diversity from published personalization research. Reviews: 7,296 analyzed via an AI review engine.

# Spotify perfected recommendation. Discovery still stalls.

~675M

monthly active users (reported)

263M

Premium subscribers

~30%

of listening is repeat / familiar

-12%

genre diversity in 6 mo · Discover Weekly (research)

## THE DISCOVERY PARADOX

Spotify owns one of the world's best recommenders — so the bottleneck was never finding new music. It's that engaged listeners won't act on it.

- **Algorithm exploits familiarity** — the more you listen, the tighter the loop
- **Unfamiliar = unrewarded** — no reason to trust a new pick, so users skip
- **Loyalty backfires** — heaviest users get the stalest discovery

## WHAT'S MISSING

Every platform competes on recommendation accuracy. None has built the acceptance layer — a reason to press play on the unfamiliar.

***The market gap is acceptance, not accuracy.***

# Everyone optimized accuracy. Nobody optimized acceptance.

## WHAT THE ECOSYSTEM SOLVED

### Recommendation accuracy

Spotify · Apple · YouTube

### Natural-language playlists

AI DJ · Prompted Playlist

### Song context / credits

SongDNA

## WHAT NOBODY SOLVED

✗ A reason to TRY the unfamiliar

✗ Trust in a brand-new rec

✗ Why THIS song, for YOU

## WHY EACH EXISTING FIX FALLS SHORT

### Discover Weekly / Daylist

Personalizes you into a tighter loop — diversity drops over time.

### AI DJ / Prompted Playlist

Lets you ask, but still just serves — generic, not personal to your history.

### SongDNA / Smart Shuffle

Facts and reshuffles — gives no reason to care about a new track.

## HYPOTHESIS

Accuracy is a solved problem. The unsolved problem is acceptance: users won't act on a recommendation they have no reason to trust.

When the unfamiliar arrives with no story and no signal, the safe move is to skip — back to the familiar.

*Users don't need better recs.  
They need a reason to say yes  
to the ones they already get.*

# We didn't survey 30 people. We analyzed 7,296 reviews.

## AI REVIEW ENGINE = RESEARCH AT SCALE

App Store · Play Store · YouTube · Community → one grounded pipeline.  
Every claim cites a real review; no fabrication.

**7,296**

reviews analyzed

**1,147**

structured-labeled (LLM)

**#1**

frustration = stale / repetitive

**1,933**

negative (from ★ ratings)

## WHAT REAL USERS SAID (CITED)

*"...it locks us into a comfortable but repetitive bubble."*

Community · #6077

*"...stuck in a sink hole surrounded by my 2020 top songs."*

Community · #6072

*"Repeating the same 30 songs out of 3000 isn't shuffle."*

App Store · #4896

## PRIMARY + SECONDARY

Planned: 5–6 interviews with Engaged Explorers (Premium power users).  
[ slot in verbatim quotes ]

Secondary research:

- Discover Weekly cut genre diversity ~12% in 6 mo
- LLMs/algorithms narrow taste as engagement rises
- Personalization → homogenization (popularity bias)

***Users aren't failing to discover — they're refusing to accept.***

# They want new music. They just won't press play on it.

## BEHAVIORAL SEGMENTATION



Target ▲

openness → · effort ↓



**Aarav G.** · 27 · Bengaluru

Premium · 4 yrs · 15 hrs/wk · 2,000+ saved songs

*“The Bubble-Trapped Explorer”*

### JOB-TO-BE-DONE

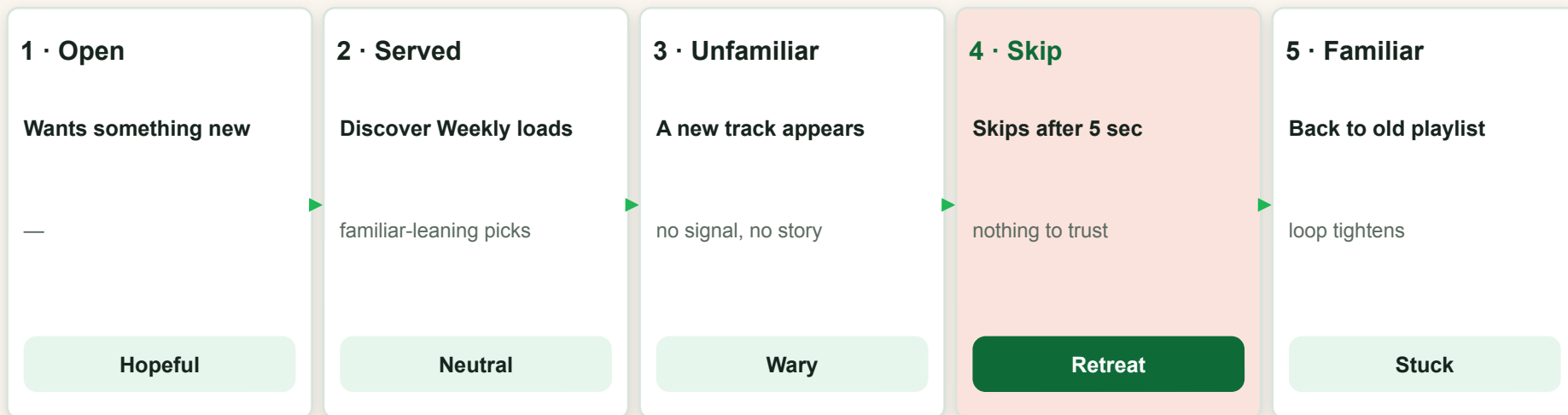
“When I open Spotify, I want to hear something genuinely new I'll love — without gambling my limited listening time on a skip.”

### PAIN POINTS

- Recs feel like a rerun of his own history
- No reason to trust an unfamiliar track
- Defaults to the same playlists, then feels stuck

*“I want new music — but everything it shows me, I've basically already heard.”*

# Discovery doesn't fail at the song. It fails at the skip.



## THE BREAKDOWN

**The failure isn't a bad recommendation — it's the silent skip at stage 4.** The unfamiliar track arrives with no story, no trust signal, no reason to risk a listen. So the user retreats, the loop tightens, and the algorithm reads the skip as 'they didn't like it' — narrowing further.

# The gap isn't accuracy. It's acceptance.

## The true problem

Recs arrive with no reason to trust them → users skip the unfamiliar and retreat to the familiar.

## Value · users

Hear genuinely new music they'll love, without gambling a skip.

## How we know

7,296 reviews: stale/repetitive = #1 discovery complaint; cross-source, cross-segment.

## Who feels it

Engaged Explorers — high-tenure Premium power users who want new music but feel trapped.

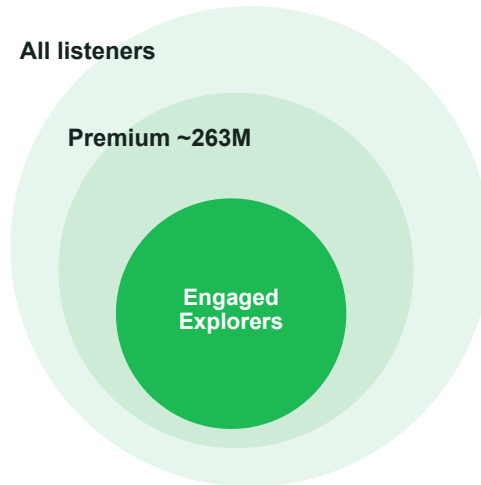
## Value · Spotify

Re-ignite discovery for the highest-LTV cohort → retention + session depth.

## Why now

Every model upgrade improves accuracy & widens the acceptance gap. No one owns it yet.

## OPPORTUNITY SIZING



Start with Engaged Explorers: highest LTV, feel the pain most, most articulate. Win them → expand to all Premium.

# Three ways to break the bubble. Only one earns trust.

## Steerable agent

### INSTRUCT

You tell discovery what you want in plain language. Powerful — but Spotify already shipped Prompted Playlist / AI DJ.

## Social vouching

### VOUCH

Show 'liked by a friend'. Strong trust — but Spotify's social graph is thin; cold-start kills it.

## Off Repeat ✓

### DECOMPOSE / PERSUADE

AI introduces each new track with a personalized story (why YOU'll like it) + trust layer. Earns acceptance.

| Basis          | Steerable | Social    | Off Repeat |
|----------------|-----------|-----------|------------|
| User insight   | 3         | 3         | 5          |
| Trust earned   | 2         | 3         | 5          |
| AI feasibility | 5         | 4         | 4          |
| Business moat  | 2         | 2         | 5          |
| <b>Total</b>   | <b>12</b> | <b>12</b> | <b>19</b>  |

## WHY OFF REPEAT WINS

- Keeps the user the decider — supports judgment, doesn't replace it
- Targets the real failure: the unfamiliar arriving with no reason to try it
- Stakes-adaptive: light for casual, rich for high-intent discovery
- Real moat: a personalized taste-narrative profile rivals can't copy



# You have On Repeat. Meet Off Repeat.

Off Repeat is the friend who puts you on. Generative AI writes a unique, grounded reason to press play for every track — impossible to template or hand-author at scale. Recommendation picks the song; AI earns the play. Trust scales in three layers:

## ♥ Real friend

“Liked by Nitya” — strongest trust

## 🎵 AI taste-twin

“Popular with your taste”

## 📖 AI story

always — works with zero friends

### ► Live prototype + workflow

[github.com/arjuncooliitr/spotify-discovery-engine](https://github.com/arjuncooliitr/spotify-discovery-engine)



## Off Repeat — for someone who loves Tame Impala + lo-fi

### Crumb — “Locket”

NEW TO YOU

psych-pop · Brooklyn

*the woozy Tame-Impala basslines you love — from a Brooklyn band you've never played.*

♥ Liked by Nitya · your friend

### Mild High Club — “Homage”

NEW TO YOU

psych-pop · Los Angeles

*lush, jazzy psychedelia — that dreamy haze you love, at a Sunday tempo.*

👤 Popular with your taste

### Men I Trust — “Show Me How”

NEW TO YOU

indie · Montreal

*the hushed lo-fi groove you keep on repeat — still way under the radar.*

✦ Picked for you

# From a stale feed to a song you'll actually try.

## SYSTEM & DATA FLOW

### 1 · Taste profile

embed listening history + saves

### 2 · Candidate retrieval

semantic match → fresh, long-tail tracks

### 3 · Novelty filter

drop anything already in the library

### 4 · Grounded narrative

LLM writes the 'why', grounded in real artist facts

### 5 · Serve + learn

trust badge + story; learns what you adopt

*The moat is the user's adoption profile — it compounds and can't be copied.*

## WHERE IT HOLDS UP UNDER PRESSURE

### Hallucinated facts

A trust feature can't lie → every story is grounded in a real artist-data source (RAG). Same no-fabrication guardrail as the engine.

### Cold start (no friends/history)

The AI story always works alone; taste-twin fills in; social is a bonus, never required.

### Story fatigue

Lean-in moments only. Default = one glanceable line; full story on tap. Silent in background listening.

# Did they discover — or just get served?

## ★ NORTH-STAR · MEANINGFUL DISCOVERY RATE

% of weekly listening on newly-adopted artists — baseline ~0 → 35% (M3) → 60% (M12). Counts only when:

### SURFACED

new to the user

### TRIED

played ≥30s

### KEPT

saved + returned in 7d

### LEADING

Story-attach try-rate · A/B story vs none

Target +25%

### BEHAVIORAL

New-artist save-rate · 7-day return

upward by M3

### GUARDRAIL

Skip-rate · session time (protect core)

flat / down

Rollout — P1 (M0–3): shadow-test on 1% of Engaged Explorers · P2 (M3–6): 10% + guardrail gates · P3 (M6–12): all Premium if discovery-rate ↑ and skip-rate flat.

*The loop becomes a launchpad — Off Repeat doesn't chase better recs, it earns the play.*

## WHERE IT CAN FAIL · WHAT WE ACCEPT

### Hallucinated story

→ ground in real facts; guardrail drops invented claims

### AI becomes a crutch

→ measure if users still explore unaided; ease off over time

### Stream-farming the reward

→ reward adoption (saved + returned), not raw plays

### Story fatigue

→ lean-in only; one glanceable line by default